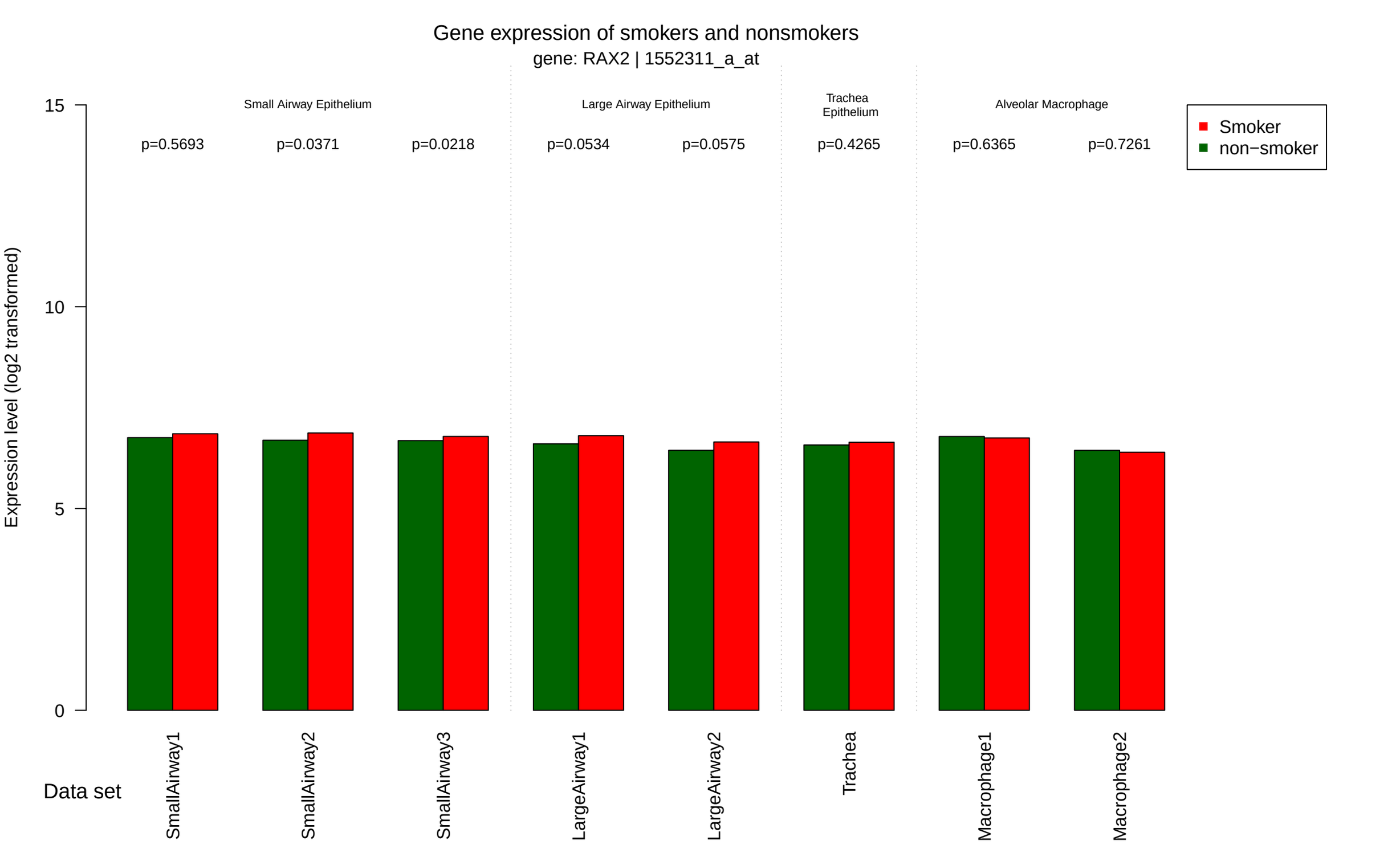
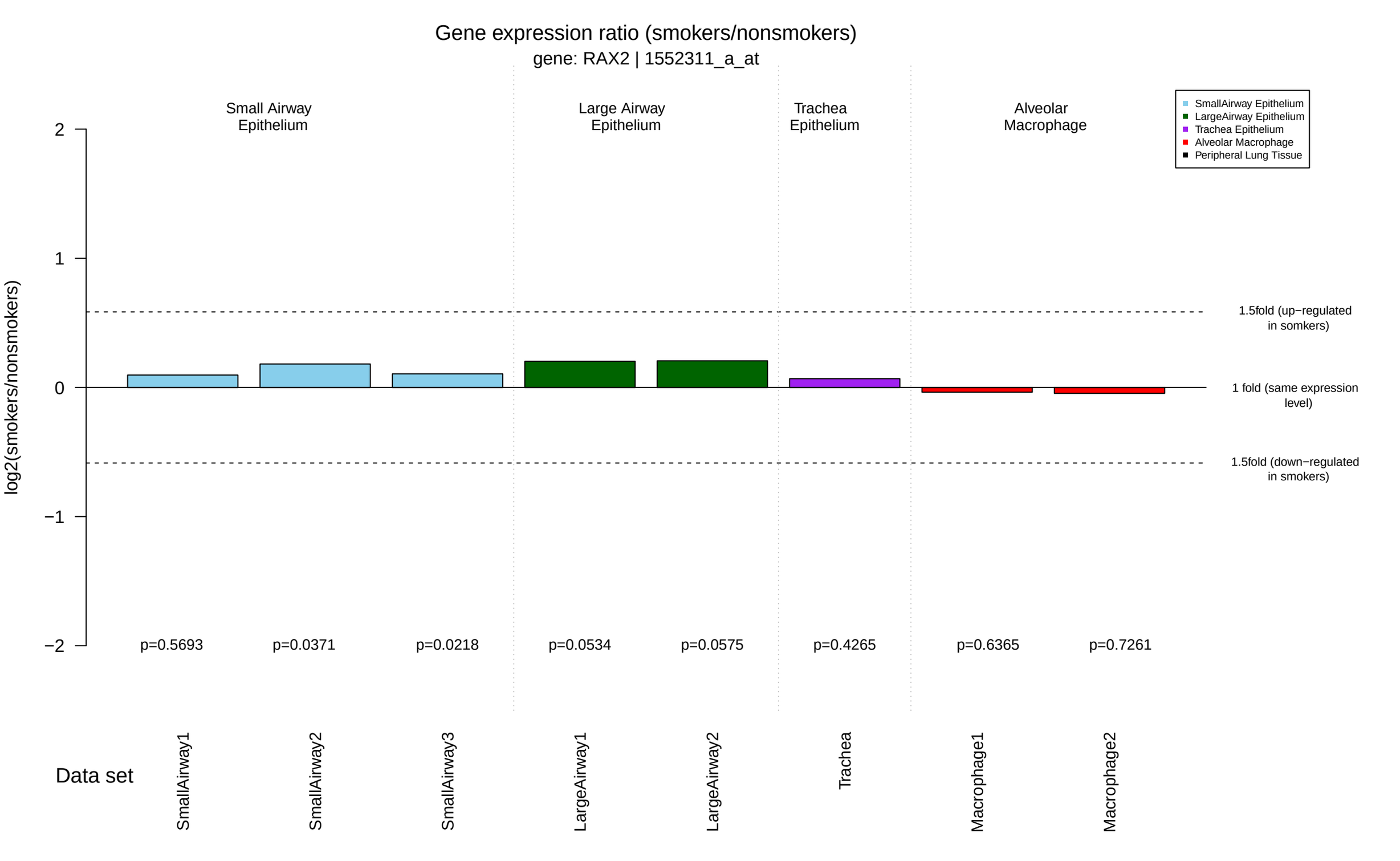
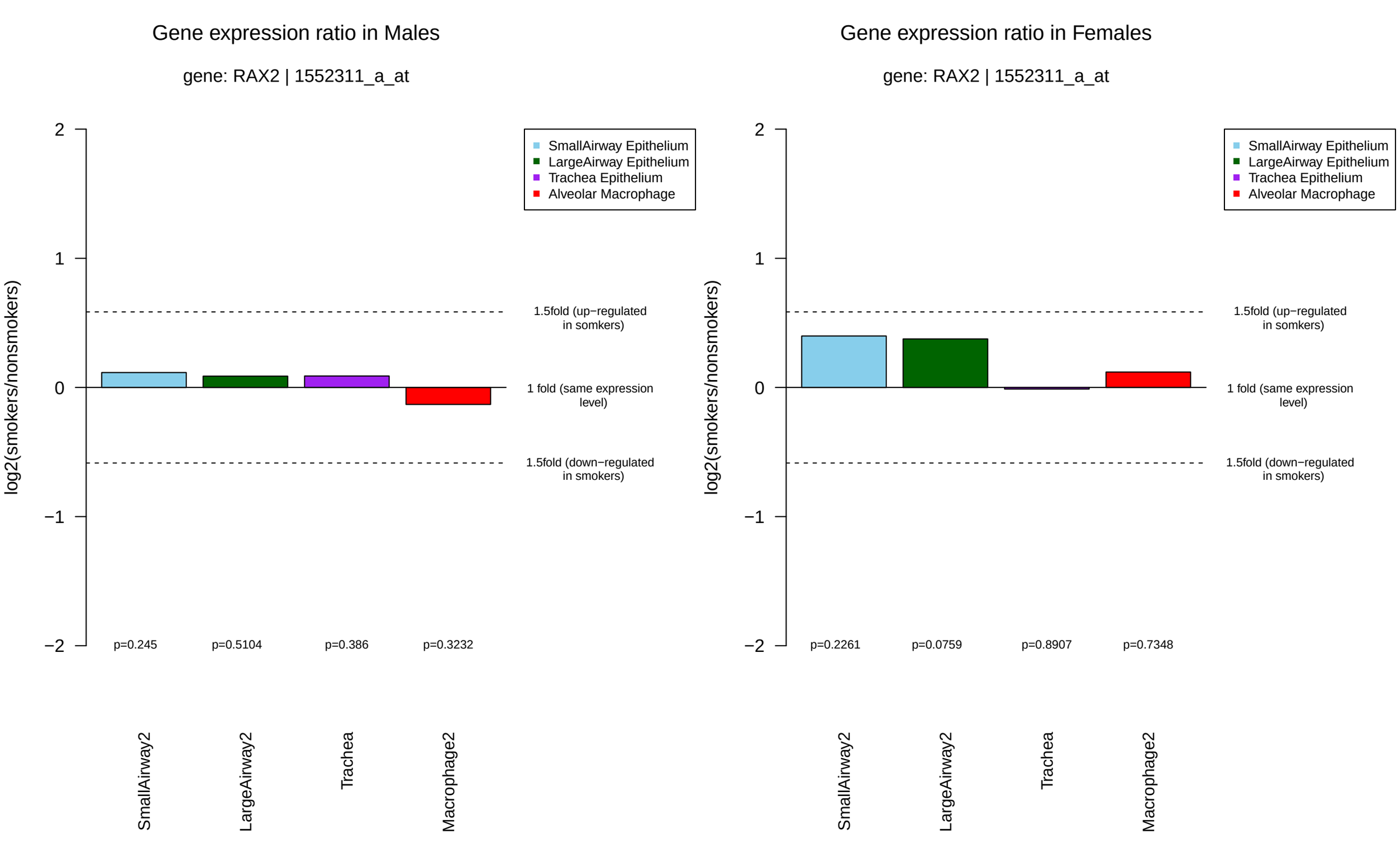
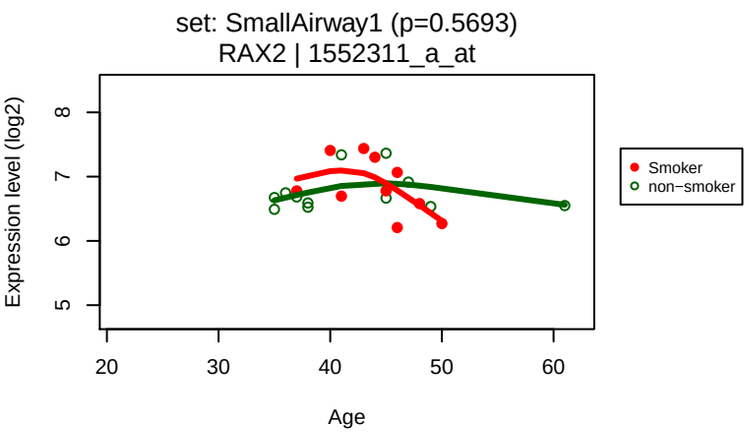
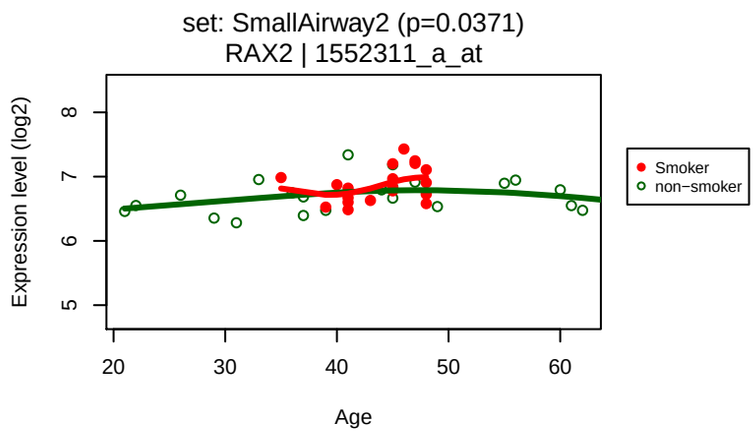
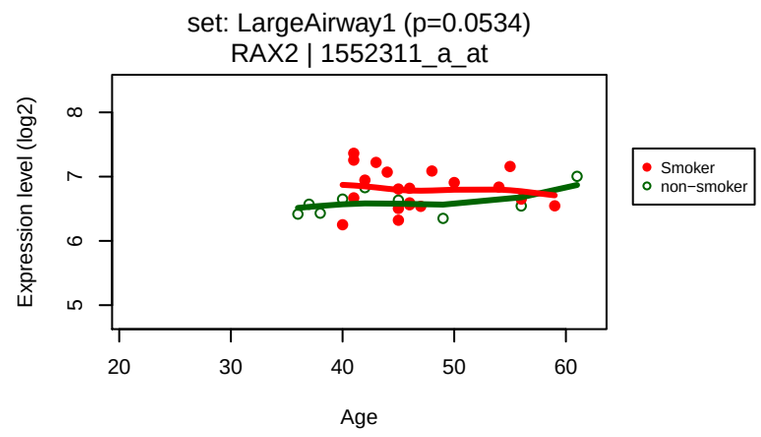

Gene expression of smokers and nonsmokers
gene: RAX2 | 1552311_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5693
p=0.0371
p=0.0218
p=0.0534
p=0.0575
p=0.4265
p=0.6365
p=0.7261
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

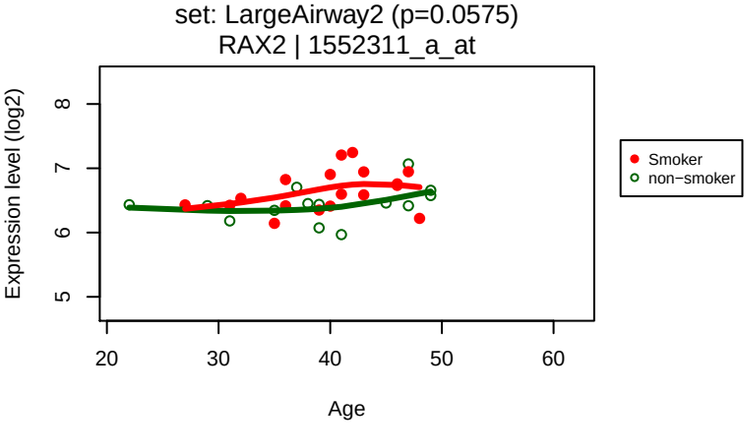

Gene expression ratio (smokers/nonsmokers)
gene: RAX2 | 1552311_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5693
p=0.0371
p=0.0218
p=0.0534
p=0.0575
p=0.4265
p=0.6365
p=0.7261
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

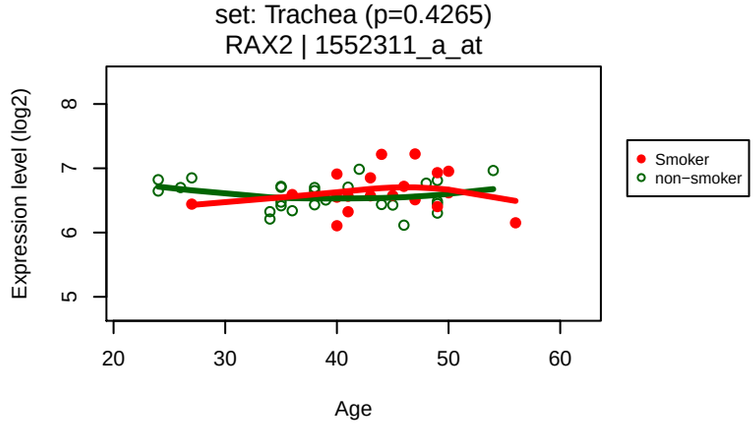

Gene expression ratio in Males
gene: RAX2 | 1552311_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.245
p=0.5104
p=0.386
p=0.3232
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAX2 | 1552311_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2261
p=0.0759
p=0.8907
p=0.7348
SmallAirway2
LargeAirway2
Trachea
Macrophage2

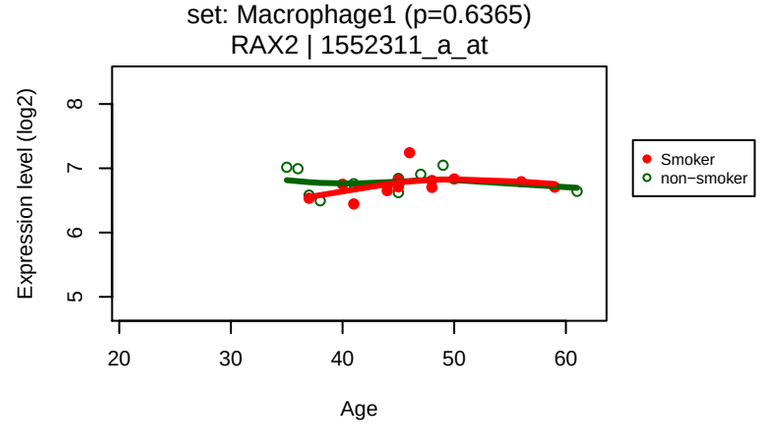

set: SmallAirway1 (p=0.5693)
RAX2 | 1552311_a_at
Expression level (log2)
Age
Smoker
non−smoker

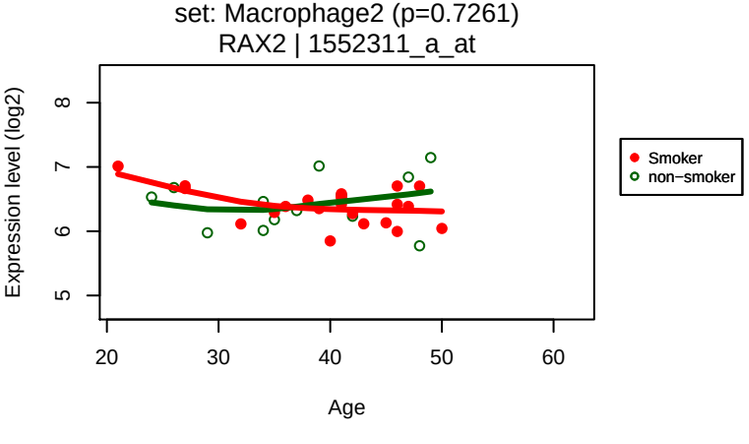

set: SmallAirway2 (p=0.0371)
RAX2 | 1552311_a_at
Expression level (log2)
Age
Smoker
non−smoker

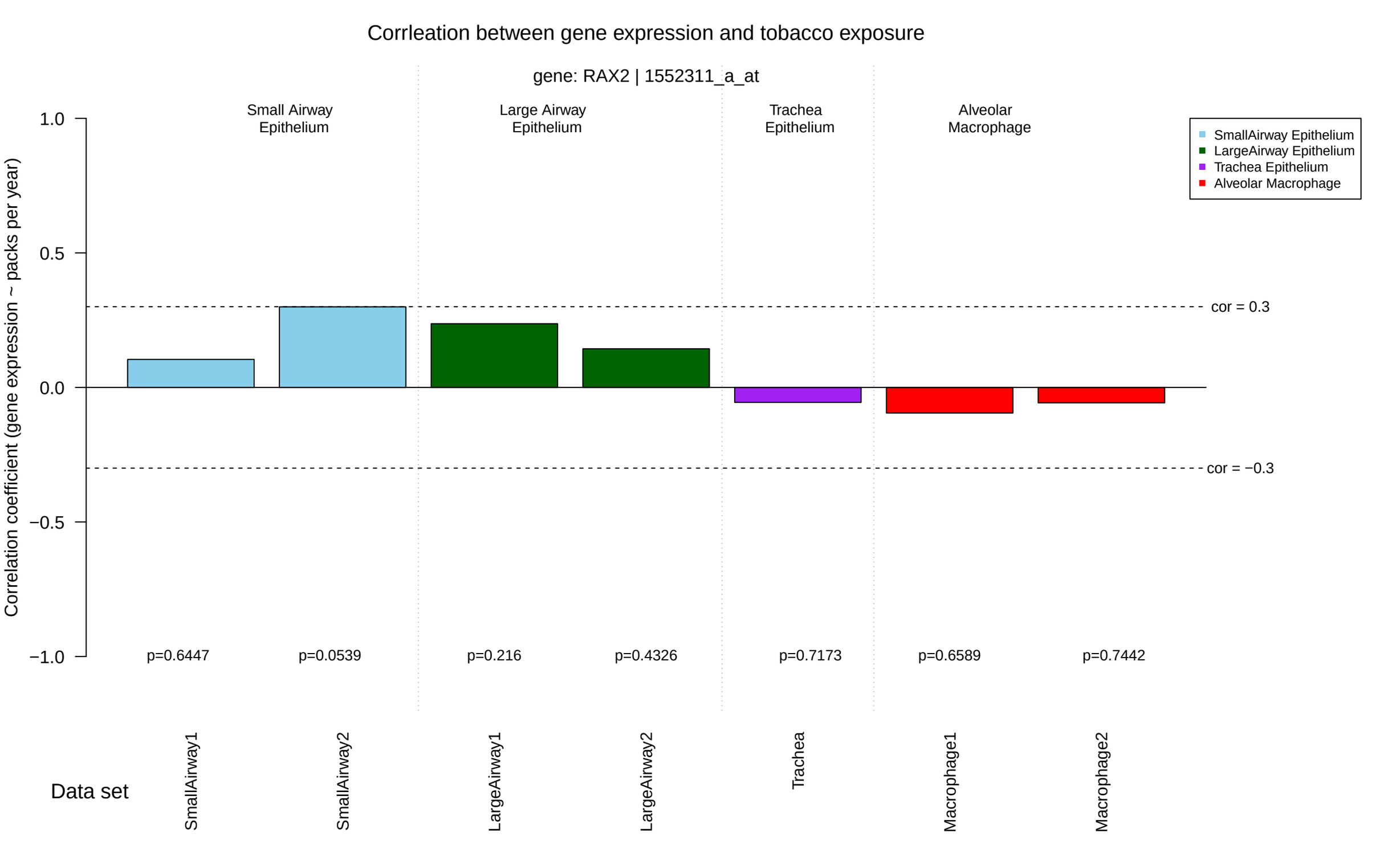

set: LargeAirway1 (p=0.0534)
RAX2 | 1552311_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0575)
RAX2 | 1552311_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4265)
RAX2 | 1552311_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6365)
RAX2 | 1552311_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7261)
RAX2 | 1552311_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAX2 | 1552311_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6447
p=0.0539
p=0.216
p=0.4326
p=0.7173
p=0.6589
p=0.7442
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2